

Project V: Advanced Software Development Tools

Assignment #2 – Debugging vs. Release Coding Practice

Purpose

The purpose of this assignment is to have you walk through the development process that will be used during the class project. While going through the development process, you will implement a set of code that will contain compiler directives for controlling debug, release, and pre-release testing code using the Visual Studio IDE and C++.

Overview

You will be implementing a basic C++ program that will read student information from an ASCII text file, store it locally in memory and display data to the screen.

Follow the instructions below careful. This assignment is about the operational flow and having your code running based on compiler directives, not based on a working application.

What to do

Step #1: Create a Project

Using Visual Studio, create a new Empty C++ solution/project in your directory of choice. Add a new source.cpp file and create a basic main function that simply returns 1.

Download a copy of the **StudentData.txt**, and **StudentData Emails.txt** files from eConestoga and add them to the root folder of your project.

Step #2: Create a Github repository

To properly demonstrate the concept of version control within the source code, you will need to create a Github repository for the assignment. Do this now, and copy a link to the repository in the box below:

<https://github.com/NimjeHarsh/Assignment2.git>

Step #3: Initial Source Code

Next you need to implement the main function to read the **StudentData.txt** file, create student objects and store them into a vector space. Create a `struct STUDENT_DATA` that contains the first and last names of the students. Each line read from the input file contains a first/last name pair separated by a comma. Parse the data, create an object and push that object into a vector space.

When you are finished you should have all the students listed in your vector space.

Commit and push your changes into the repository master.

Step #4: Adding in some `DEBUG` functionality

To help debug the loading of the file update the main function to print out all the student information to the screen. This functionality should only be performed if the application has been compiled in debug mode.

Commit and push your changes into the repository master.

Step #5: Creating a Pre-Release

Your next task is to add new functionality into the software, but in a way where it can be merged, but not executed unless running in a pre-release executable. This new functionality will add email addresses to your student input data. Use the following steps to create a pre-release version of your source code:

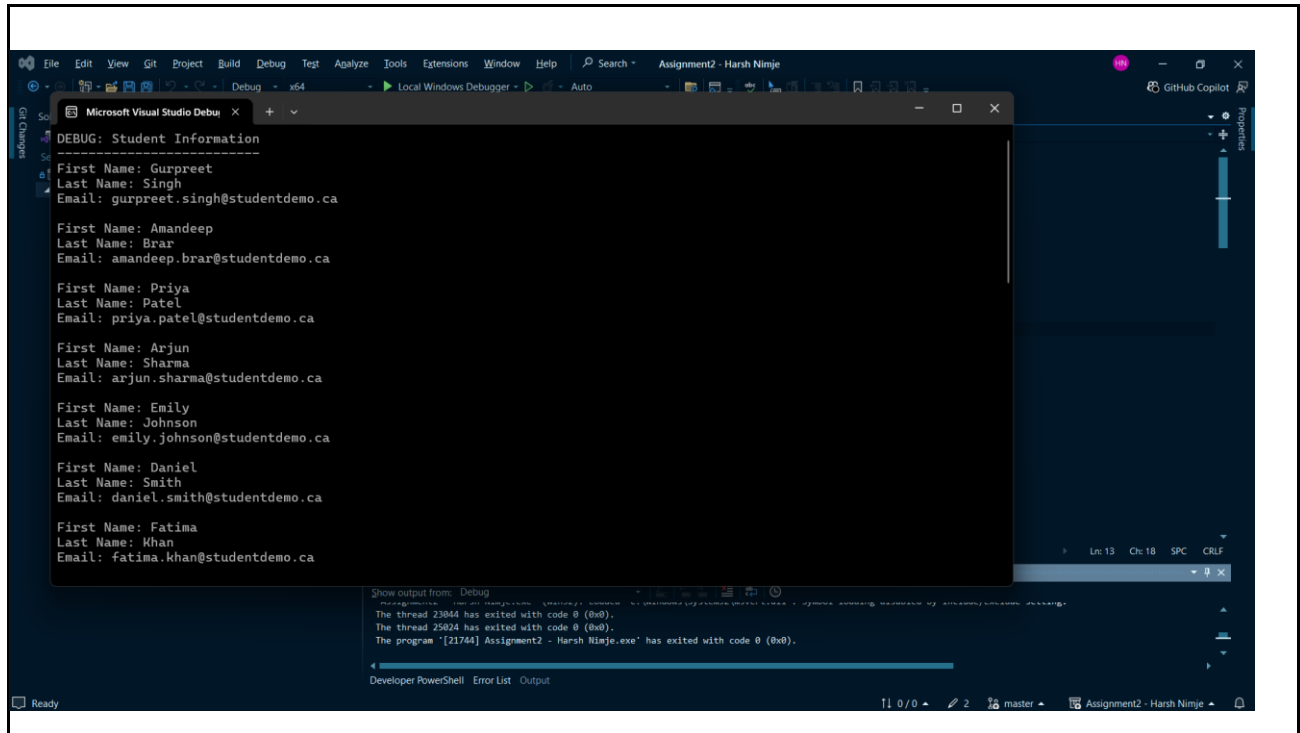
1. Create a branch in Github called **PreRelease**, commit and push it into your Github Repo
2. Update the main source code to print out a message to state if the application is running standard or pre-release source code
3. Update the main source code to read the **StudentData_Emails.txt** if compiled using a `PRE_RELEASE` compiler directive, commit and push into your Github Repo PreRelease branch
4. Perform a "Pull Request" on your **PreRelease** branch
 - a. Merge your PreRelease branch code into your master branch
5. In Visual Studio, resync your project with Github.

Step #6: Demo and Execution

For the final step you are going to execute and screenshot your application running in the different modes.

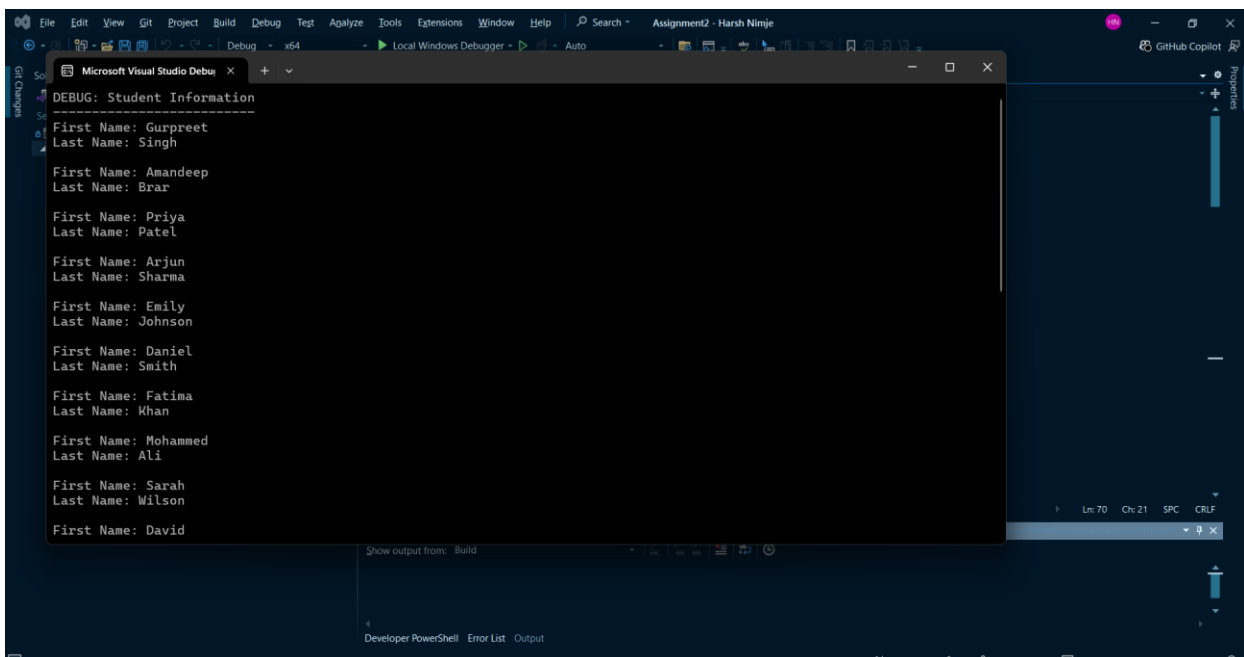
Screenshot #1: `DEBUG` – Pre-Release Version

Set visual studio to "debug" and enable your PreRelease compiler directives. Compile and run your program. Take a screenshot of the console window and past it below:



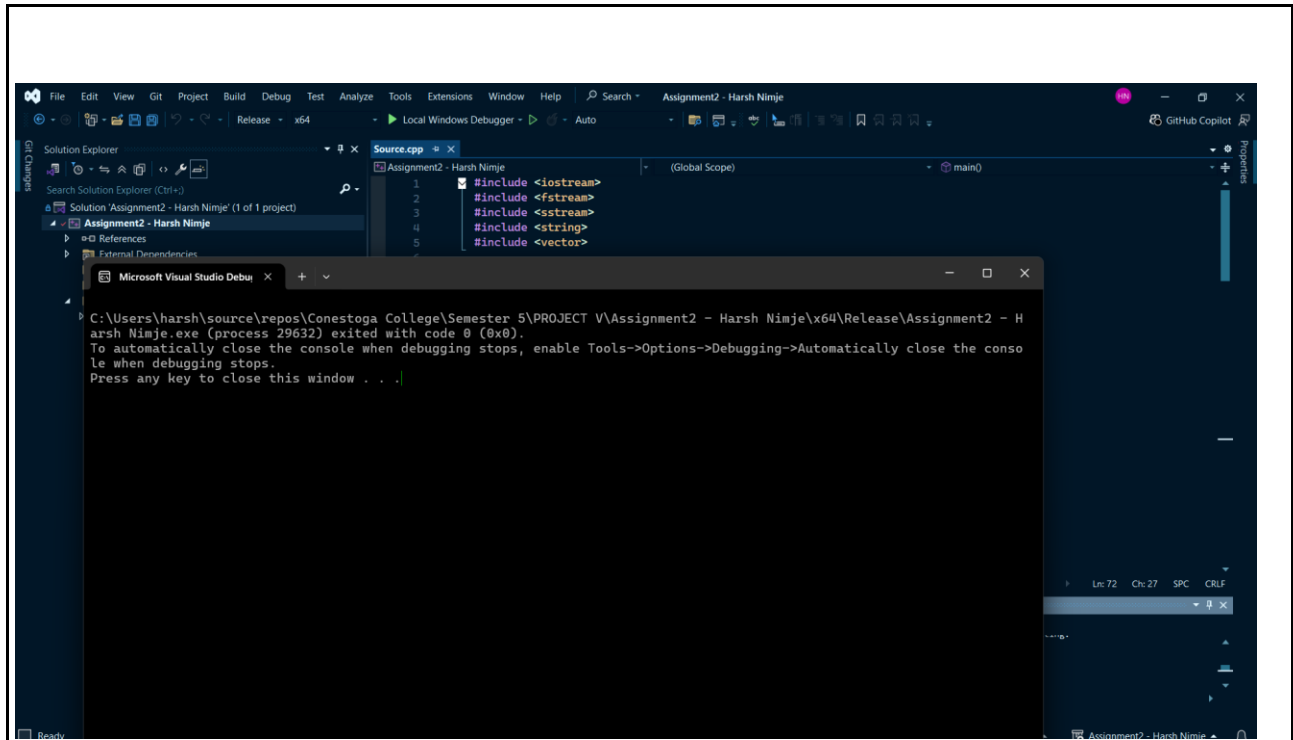
Screenshot #2: DEBUG – Standard Version

Set visual studio to “debug” and disable your PreRelease compiler directives. Compile and run your program. Take a screenshot of the console window and paste it below:



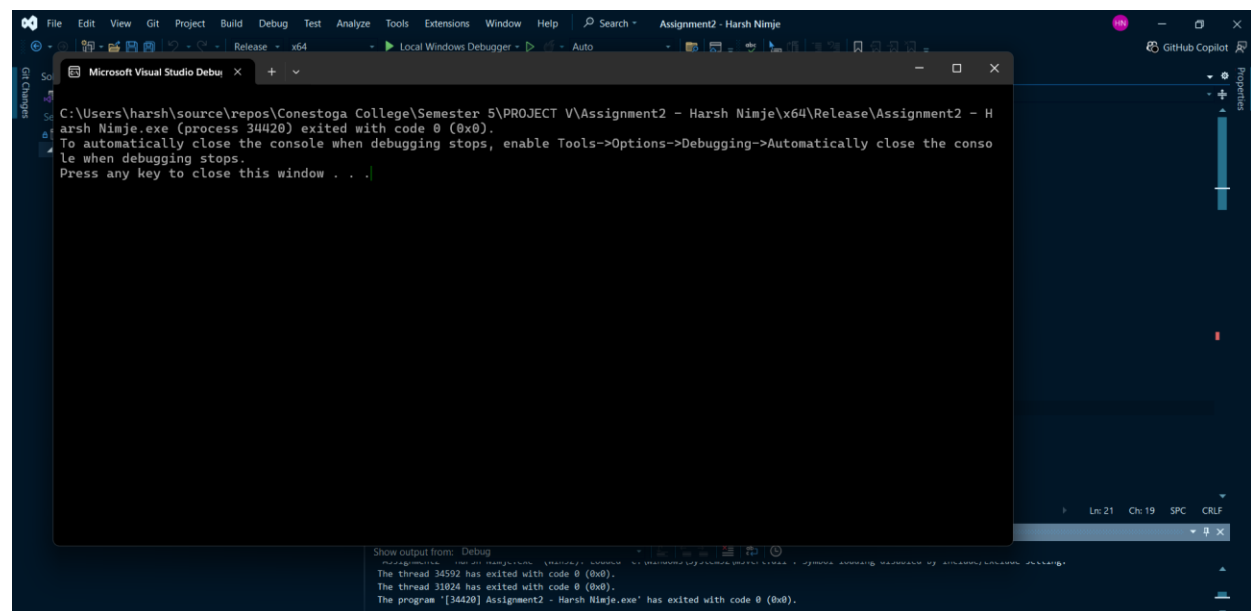
Screenshot #3: RELEASE – Pre-Release Version

Set visual studio to “release” and enable your PreRelease compiler directives. Compile and run your program. Take a screenshot of the console window and paste it below:



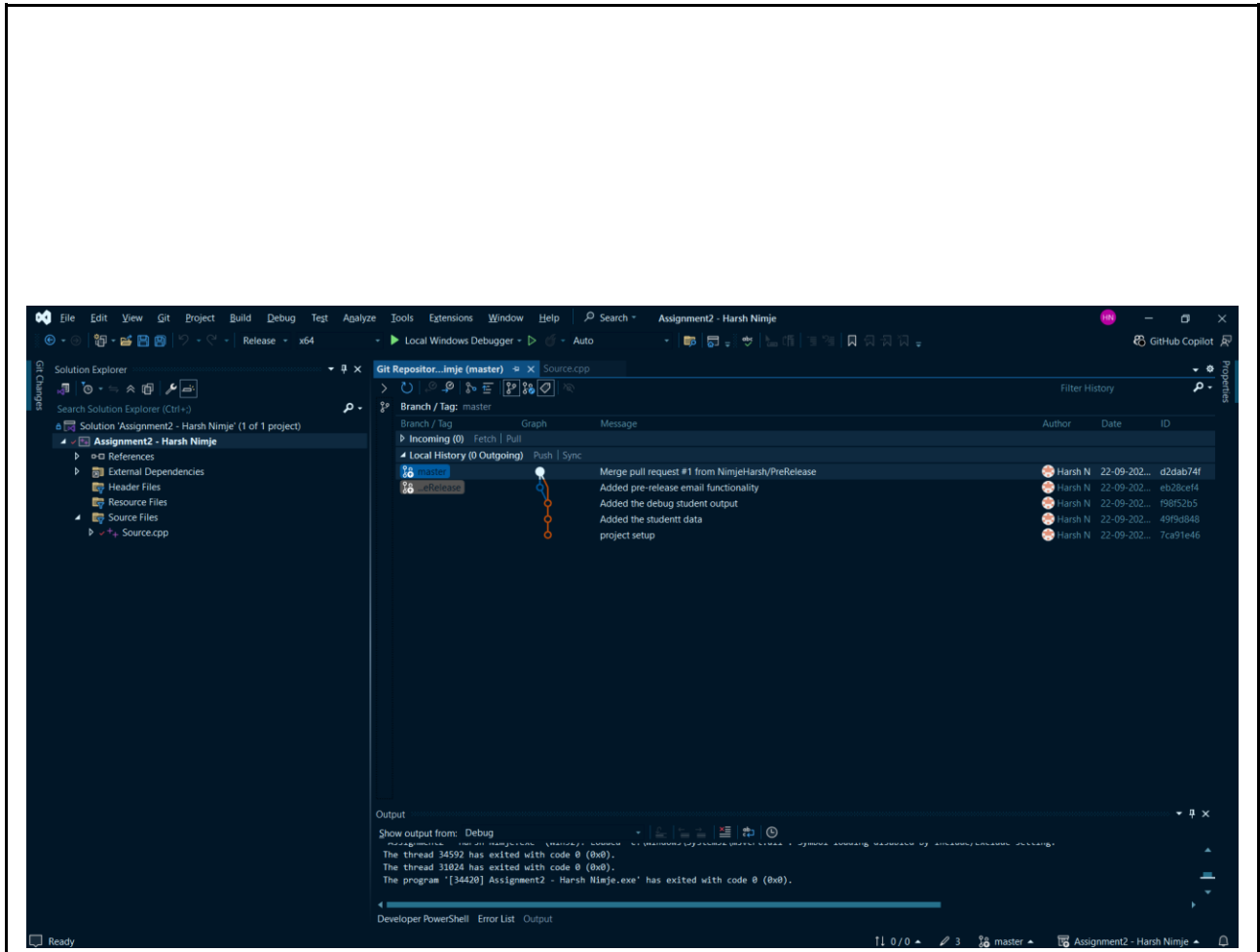
Screenshot #4: RELEASE – Standard Version

Set visual studio to “release” and disable your PreRelease compiler directives. Compile and run your program. Take a screenshot of the console window and paste it below:



Screenshot #5: Visual Studio Branch History

In visual studio select “Git → Manage Branches” from the main menu bar. Take a screenshot of your visual studio with the “Git Repository” tab open and place it in the box below:



Rubric

See eConestoga.

What to Hand In

Please submit the following files to eConestoga for marking

- A completed copy of this PDF document
- A ZIP of your Visual Studio solution space **Please delete all "RELEASE" and "DEBUG" directories.**